

Dax L. Feliz, PhD | Curriculum Vitae



Last updated: April 1, 2026

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Research Interests

Developing an end-to-end observational program to forecast atmosphere retention in nearby M-dwarf planetary systems, combining: (1) completeness-controlled transit surveys of M-dwarfs in TESS FFIs (NEMESIS); (2) multi-epoch flare frequency distributions and stellar activity evolution from TESS, K2, and ground-based photometry; and (3) Galactic-context characterization via Gaia kinematics and SDSS spectroscopic metallicities. Long-term goal: a ranked, observation-ready target list for atmospheric characterization with JWST, Roman, and PLATO.



Academic Appointments

Flatiron Research Fellow, Flatiron Center for Computational Astrophysics (CCA), September 2024 – Present

Postdoctoral Advisor: Dr. David Hogg

Developing an independent research program measuring exoplanet occurrence rates and flare frequency distributions around nearby M-dwarfs using NASA TESS and ground-based photometry; extending the NEMESIS pipeline for FFI-based transit surveys; initiated and co-supervised one CUNY Bridge graduate thesis project and one undergraduate research project.

Postdoctoral Researcher, American Museum of Natural History (AMNH), September 2022 – September 2024

Postdoctoral Advisor: Dr. Ruth Angus

Led independent transit and stellar activity analyses of nearby M-dwarfs using TESS Full Frame Images photometry; initiated and co-supervised one CUNY Bridge graduate thesis project, two undergraduate internship projects and six high school internship projects.

Postdoctoral Researcher, Vanderbilt University, May 2022 - August 2022

Postdoctoral Advisor: Dr. Keivan Stassun

Visiting Graduate Student Fellow, George Mason University, 2021 - 2022

Advisors: Drs. Peter Plavchan & Keivan Stassun



Education

PhD Astrophysics, Vanderbilt University (VU), May 2022. Advisor: Dr. Keivan Stassun,
[Dissertation Link](#)

M.A. Physics, Fisk University, May 2018. Advisors: Drs. Keivan Stassun and Karen Collins

B.S. Astrophysics, University of Massachusetts at Amherst, May 2013. Advisor: Dr. Min Yun



Selected Research Contributions / Research Program

Atmosphere Retention Forecasting for Nearby M-dwarf Planets, CCA 2025 – Present

Developing a predictive Atmosphere Retention Metric (ARM) that combines per-planet escape velocity with star-by-star historic XUV exposure histories inferred from completeness-controlled flare frequency distributions, rotation periods, and Gaia/SDSS Galactic context. Outputs a ranked atmospheric characterization target list for JWST, Roman, and PLATO follow-up.

Transit Survey of M-Dwarf stars in NASA TESS data, VU/AMNH/CCA 2018 – Present

PhD Thesis: Built NEMESIS pipeline to extract NASA TESS Full Frame Image light curves, remove instrumental systematics, and run periodic transit searches to surface high-value candidates.

ESA PLATO Flare Removal Working Group, European Space Agency, 2020 – Present

Constructing an algorithm to detect and remove flare events from the photometric time series data that will be produced by the ESA PLATO mission.

Kilodegree Extremely Little Telescope (KELT), VU, TN 2017 – 2022

Operating KELT data reduction pipeline for detection of transiting exoplanets orbiting bright, hot stars with a network of ground-based telescopes as a collaboration between Vanderbilt University and The Ohio State University.

Multi-Year Transit Survey of Proxima Centauri, Fisk University and VU, TN 2016 – 2019

Master's Thesis Project: Processing images from observations of the star Proxima Centauri, production and detrending analysis of light curves and utilization of periodic transit search algorithms to hunt for transit events of the Earth-sized planet, Proxima b.

**Publication Metrics**

Refereed: 31 / 1st Author: 3 / Citations: 1,007 / h-index: 17 (2026-02-24, via ADS).

**Fellowships**

NSF AGEP Fellowship, Spring 2021 – 2022. \$57,272 (*competitive national award*)

Fisk-Vanderbilt Bridge Fellowship, Fall 2018 – 2021. \$39,000 annual stipend.

**Successful Proposals / Grants**

NASA Transiting Exoplanet Survey Satellite (TESS) Guest Investigator Cycle 9, *Long-Term Evolution of M-dwarf Flare Activity with K2, TESS, and Ground-Based Photometry* – PI: **D. L. Feliz**. Submitted March 2026. (*NASA ROSES; under review*)

NSF Graduate Opportunities at Fisk in Astronomy and Astrophysics Research (GO-FAAR) grant, 2016 – 2018. NSF Grant #1358862. \$20,000 annual stipend.

McMinn Summer Research Award, 2018. \$5,000

New York Space Consortium Research Grant, Summer of 2012. \$5,000

**Honors & Recognition**

Heroes Classroom Namesake: The Feliz Room, East Harlem Scholars Academy, New York, NY, 2013.

**Professional Service**

Member, ESA PLATO Flare Removal Working Group, 2020 – Present

Instructor, CUNY Master's in Astrophysics Bridge Programming Bootcamp, 2022 – Present

Curriculum developer and lead instructor for annual two-week graduate onboarding bootcamp; ~10–15 students per cohort.

**Invited Colloquia & Seminars** (*4 invited / 2 contributed colloquia & seminars*)

- 'From Exoplanet Detections to Atmosphere Forecasts Around Nearby M-Dwarfs', New York City College of Technology Physics Departmental Seminar, February 2026.

- "NEMESIS I: Exoplanet Transit Survey of Nearby M-dwarfs in TESS FFIS I" - Presented at:

- MIT TESS Science Talk (Mar 2021)

- Rubin LSST TVS Seminar (Jun 2021)

- Center for Exoplanets & Habitable Worlds Seminar, Penn State (Sep 2021)

- "How to deflect an incoming asteroid", University of Massachusetts at Amherst Astronomy Colloquium, (May 2013).

**Invited Conference & Workshop Talks** (*7 invited conference & workshop talks*)

- "NEMESIS II: Exoplanet Transit Survey of Nearby M-dwarfs in TESS FFIs" - Presented at:

- Emerging Researchers in Exoplanets (ERES) X, Princeton University (June 2025)

- "From Trends to Transits" Workshop, University of New Mexico (Aug 2025)
- 'M-dwarfs are cool!', AstroFest at the Flatiron Institute Center for Computational Astrophysics (September 2024 & September 2025)
- "Twinkle, Twinkle Little Star", Twinkle and the Next Generation of Exoplanet Scientists Conference (Sep 2021).
- "Using BLS modeling on photometry of Proxima Centauri", KELT Workshop, Lehigh University (Jun 2017).
- "Light Curve Characterization of High Proper Motion M-Dwarf Stars with the Kepler Space Telescope", AMNH REU Annual Symposium (Aug 2012).
- "Using Spectropolarimetry to Study Extrasolar Planets", AMNH REU Annual Symposium (Aug 2010)



Contributed Talks & Posters (5 contributed talks & posters)

- "NEMESIS II: Exoplanet Transit Survey of Nearby M-dwarfs in TESS FFIs", Presented at:
 - TESS Science Conference III, Massachusetts Institute of Technology (Jul 2024, <https://zenodo.org/records/12973447>)
 - Sagan Summer Workshop, California Institute of Technology (Jul 2025, https://nexsci.caltech.edu/workshop/2025/posters/Poster_DaxFeliz_101.pdf)
- "Can Photometric Flares Detected by Ground and Space Based Telescopes Be Used To Identify Activity Cycles in M-dwarf stars?", Cool Stars 22 (Jul 2024; poster <https://zenodo.org/records/13146889>)
- "NEMESIS I: Exoplanet Transit Survey of Nearby M-dwarfs in TESS FFIS I", Presented at:
 - Cool Stars 20.5 (Mar 2021; poster <http://doi.org/10.5281/zenodo.4562794>)
 - TESS Science Conference II (Aug 2021; <http://doi.org/10.5281/zenodo.5115302>)



Mentoring & Advising Experience

Research mentoring across high school, undergraduate, and graduate levels with projects in exoplanet detection, stellar activity, and time-series modeling; trainees have presented at conferences and workshops, and have advanced to graduate programs and industry jobs.

Student Project Totals by Category: High School Students: 7 ; College Students: 5 ; Graduate Students: 2.

CUNY Master's in Astrophysics, Bridge Thesis Project

Searching for M dwarf Coronal Mass Ejections with TESS and SDSS-V, 2024 – *Present*. Graduate Student: Andrea Bracamonte, City University of New York Graduate Center, Co-Advised with Dr. Ruth Angus. Project outputs contribute a framework for producing flare-rate priors and SDSS spectroscopic detection efficiency estimates for the NEMESIS atmosphere-retention forecasting program.

Barnard Summer Research Institute AMNH Project

Detection and Validation of Transiting Exoplanets Around Nearby M-dwarf Stars, May 2024 - August 2024. Graduate Student: Madeline J. Maldonado Gutierrez, Barnard College of Columbia University.

CUNY Master's in Astrophysics, Bridge Thesis Project

Using Gaussian Processes to model stellar rotation in Zwicky Transient Facility photometry, 2022 - 2024. Graduate Student: Ryan Lebron, City University of New York Graduate Center, Co-Advised with Dr. Ruth Angus. Project outputs contribute GP-derived rotation period catalog used as stellar activity priors in the NEMESIS atmosphere-retention forecasting program.

AMNH Research Experience for Undergraduates (REU) Program

Detecting Exoplanets Around Nearby M-dwarf Stars from the Revised TESS Habitable Zone Catalog, 2023. Undergraduate Student: Maliyah Adams, Arizona State University

AMNH Science Research Mentoring Program

Detection of Transiting Exoplanets and Eclipsing Binaries Around Nearby M-dwarf Stars, 2022 - 2024. High School Students: August Fischer, Donovan Bradley, Jashcelyn Canada (2022 – 2023), Kylor Ghai, Shreeya KC, Thamim Chowdhury (2023-2024)

Central American-Caribbean Bridge Program

Project: Recovery of known TESS Objects of Interest from Full Frame Images of TESS observations, 2020 - 2021. Undergraduate Student: Bryan Villarreal Alvarado, University of Costa Rica.

Summer Research Internship

Project: Transiting Exoplanet Survey Satellite: A Search for New Worlds, 2019 - 2020.

Undergraduate Student: Samantha Bianco, Vanderbilt University.

Summer Research Internship

Project: Blind Transit Survey of TESS Data for M Dwarf Systems, 2019 - 2020.

Undergraduate Student: Mary Jimenez, George Mason University.

School for Science and Math at Vanderbilt

Project: Detecting exoplanet transits in TESS light curves, Spring 2019.

High School Student: Felix Bean, Hunter's Lane High School.



Teaching Experience

CUNY Master's in Astrophysics Bridge Program

 **Programming Bootcamp Instructor**, August 2022, 2023, 2024 & 2025.

A two-week orientation and programming bootcamp for incoming astronomy graduate students of varied backgrounds; ~10–15 students per cohort. Responsibilities include curriculum design and development, instruction in Python, data analysis, and astronomical software tools, and integration of students into the NYC astronomy community. Course materials publicly available at <https://daxfeliz.github.io/cunybridgebootcamp/>.

Teaching Assistantships

ASTR-101L: Introductory Solar System Laboratory, Vanderbilt University, TN Fall semesters, 2018 – 2020.

ASTR-102L: Introductory Stars & Galaxies Laboratory, Vanderbilt University, TN Spring semesters, 2018 – Spring 2020.

Laboratory Instruction / Teaching Support

Electronics Laboratory, Bronx Community College, NY Spring 2014 – Fall, 2015.



Software and Open Science Products

Research Software & Pipelines (Selected):

NEMESIS (TESS Full Frame Image transit-photometry pipeline) – Developer (2018 – Present)

Built pipeline to extract TESS FFI light curves, remove instrumental systematics, and run periodic transit searches to identify candidates. Pipeline scope extended in to include multi-epoch stellar flare detection, flare frequency distribution inference, and Galactic-context characterization via Gaia and SDSS cross-matching. *Status*: Built, in development.

ESA PLATO flare-removal algorithm (PLATO Flares Working Group) – Developer/Contributor (2020 – Present)

Developing methods to detect and remove stellar flare events in PLATO mission photometric time series. *Status*: Built, in development.

Kilodegree Extremely Little Telescope data reduction pipeline – Operator/Analyst (2017 – 2022)

Operated pipeline for transit detections in bright-star ground-based survey data. *Status*: operated/main-tained/applied

Data Products / Catalogs / Releases (Selected):

NEMESIS I Planet Candidates

Catalog of vetted Community TESS Objects of Interest from the NEMESIS I blind-transit survey of TESS Full Frame Images. URL: <https://filtergraph.com/NEMESIS>

Computational Training Materials:

Programming Bootcamp tutorials for CUNY Master's Program in Astrophysics Program

URL: <https://daxfeliz.github.io/cunybridgebootcamp/>



Outreach, & Public Engagement

East Harlem Scholars Academy, New York, NY, Alumni Day Speaker, 2010 – 2015

- Answering questions of elementary school students about astronomy and physics and my research experience.

Central Park East High School, New York, NY, College Panel Speaker, 2009 – 2013

- Answering questions of high-school students about my experiences in science and in higher education.

Central Park East High School, New York, NY, Robotics Consultant, 2009 – 2011

- Volunteered to assist high school students in the FIRST Robotics Competition.

East Harlem Tutorial Program, New York, NY, 2008 – 2010.

- Tutored and mentored many high-school and middle school students in math, science and the college application process. I was also a mentor in the For Inspiration and Recognition of Science and Technology (FIRST) robotics competition.



First-Author Refereed Publications

1. **Dax L. Feliz**, Peter Plavchan, Samantha N. Bianco, and ... NEMESIS: Exoplanet Transit Survey of Nearby M-dwarfs in TESS FFIs. I. *The Astronomical Journal*, 161(5):247, May 2021
2. **Dax L. Feliz** L., David L. Blank, Karen A. Collins, and ... A Multi-year Search for Transits of Proxima Centauri. II. No Evidence for Transit Events with Periods between 1 and 30 days. *The Astronomical Journal*, 157(6):226, Jun 2019
3. David L. Blank, **Dax L. Feliz**^{**}, Karen A. Collins, and ... A Multi-year Search for Transits of Proxima Centauri. I. Light Curves Corresponding to Published Ephemerides. *The Astronomical Journal*, 155(6):228, Jun 2018, ^{**Corresponding author}



Key Collaborative Publications

1. Peter Plavchan, Thomas Barclay, Jonathan Gagné, ... **Dax Feliz**, and ... A planet within the debris disk around the pre-main-sequence star AU Microscopii. *Nature*, 582(7813):497–500, June 2020
2. Heike Rauer, Conny Aerts, Cabrera, ..., **D. L. Feliz**, and ... The PLATO mission. *Experimental Astronomy*, 59(3):26, June 2025
3. Karen A. Collins, Kevin I. Collins, Joshua Pepper, ..., **Dax L. Feliz**, and ... The KELT Follow-up Network and Transit False-positive Catalog: Pre-vetted False Positives for TESS. *The Astronomical Journal*, 156(5):234, Nov 2018



Peer Reviewed Publications (Nth Author)

1. Madison G. Scott, Georgina Dransfield, Mathilde Timmermans, ..., **Dax L. Feliz**, and ... Two temperate Earth- and Neptune-sized planets orbiting fully convective M dwarfs. *Monthly Notices of the Royal Astronomical Society*, January 2026
2. N. Heidari, G. Hébrard, E. Martioli, ..., **D. L. Feliz**, and ... Characterization of seven transiting systems, including four warm Jupiters from SOPHIE and TESS. *Astronomy & Astrophysics*, 694:A36, February 2025
3. E. Semenko, O. Kochukhov, Z. Mikulášek, ..., **D. L. Feliz**, and ... HD 34736: an intensely magnetised double-lined spectroscopic binary with rapidly rotating chemically peculiar B-type components. *Monthly Notices of the Royal Astronomical Society*, 535(3):2812–2836, December 2024
4. Rishi R. Paudel, Thomas Barclay, Allison Youngblood, ..., **Dax L. Feliz**, and ... A Multiwavelength Survey of Nearby M Dwarfs: Optical and Near-ultraviolet Flares and Activity with Contemporaneous TESS, Kepler/K2, Swift, and HST Observations. *The Astrophysical Journal*, 971(1):24, August 2024
5. Lionel J. Garcia, Daniel Foreman-Mackey, Catriona A. Murray, ..., **Dax L. Feliz**, and ... nuance: Efficient Detection of Planets Transiting Active Stars. *The Astronomical Journal*, 167(6):284, June 2024
6. Christopher R. Mann, Paul A. Dalba, David Lafrenière, ..., **Dax L. Feliz**, and ... Giant Outer Transiting Exoplanet Mass (GOT 'EM) Survey. III. Recovery and Confirmation of a Temperate, Mildly Eccentric, Single-transit Jupiter Orbiting TOI-2010. *The Astronomical Journal*, 166(6):239, December 2023

7. Justin M. Wittrock, Peter P. Plavchan, Bryson L. Cale, ..., **Dax L. Feliz**, and ... Validating AU Microscopii d with Transit Timing Variations. *The Astronomical Journal*, 166(6):232, December 2023
8. Ismael Mireles, Diana Dragomir, Hugh P. Osborn, ..., **Dax L. Feliz**, and ... TOI-4600 b and c: Two Long-period Giant Planets Orbiting an Early K Dwarf. *The Astrophysical Journal Letters*, 954(1):L15, September 2023
9. Keivan G. Stassun, Guillermo Torres, Marina Kounkel, ..., **Dax L. Feliz**, and ... An Eclipsing Binary Comprising Two Active Red Stragglers of Identical Mass and Synchronized Rotation: A Post-mass-transfer System or Just Born That Way? *The Astrophysical Journal*, 950(2):99, June 2023
10. Joseph E. Rodriguez, Samuel N. Quinn, Andrew Vanderburg, ..., **Dax L. Feliz**, and ... Another Shipment of Six Short-Period Giant Planets from TESS. *Monthly Notices of the Royal Astronomical Society*, February 2023
11. Mohammed El Mufti, Peter P. Plavchan, Howard Isaacson, ..., **Dax L. Feliz**, and ... TOI 560: Two Transiting Planets Orbiting a K Dwarf Validated with iSHELL, PFS, and HIRES RVs. *The Astronomical Journal*, 165(1):10, January 2023
12. Keivan G. Stassun, Guillermo Torres, ..., **Dax L. Feliz**, and ... A Low-mass Pre-main-sequence Eclipsing Binary in Lower Centaurus Crux Discovered with TESS. *The Astrophysical Journal*, 941(2):125, December 2022
13. S. Ulmer-Moll, M. Lendl, S. Gill, ..., **D. L. Feliz**, and ... Two long-period transiting exoplanets on eccentric orbits: NGTS-20 b (TOI-5152 b) and TOI-5153 b. *arXiv e-prints*, page arXiv:2207.03911, July 2022
14. Ian Stotesbury, Billy Edwards, ..., **Dax L. Feliz**, and ... Twinkle: a small satellite spectroscopy mission for the next phase of exoplanet science. In Laura E. Coyle, Shuji Matsuura, and Marshall D. Perrin, editors, *Space Telescopes and Instrumentation 2022: Optical, Infrared, and Millimeter Wave*, volume 12180 of *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series*, page 1218033, August 2022
15. Justin M. Wittrock, Stefan Dreizler, Michael A. Reefer, Brett M. Morris, ..., **D. L. Feliz**, and ... Transit Timing Variations for AU Microscopii b and c. *The Astronomical Journal*, 164(1):27, July 2022
16. Michael A. Reefer, Rafael Luque, Eric Gaidos, ..., **D. L. Feliz**, and ... A Close-in Puffy Neptune with Hidden Friends: The Enigma of TOI 620. *The Astronomical Journal*, 163(6):269, June 2022
17. Jiayin Dong, Chelsea X. Huang, George Zhou, ..., **D. L. Feliz**, and ... NEID Rossiter-McLaughlin Measurement of TOI-1268b: A Young Warm Saturn Aligned with Its Cool Host Star. *The Astrophysical Journal Letters*, 926(2):L7, February 2022
18. Christina Hedges, Alex Hughes, George Zhou, ..., **D. L. Feliz**, and ... TOI-2076 and TOI-1807: Two Young, Comoving Planetary Systems within 50 pc Identified by TESS that are Ideal Candidates for Further Follow Up. *The Astronomical Journal*, 162(2):54, August 2021
19. Keivan G. Stassun, Guillermo Torres, Cole Johnston, ..., **Dax L. Feliz**, and ... Discovery and Characterization of a Rare Magnetic Hybrid β Cephei Slowly Pulsating B-type Star in an Eclipsing Binary in the Young Open Cluster NGC 6193. *The Astrophysical Journal*, 910(2):133, April 2021
20. S. A. Rappaport, D. W. Kurtz, G. Handler, ..., **D. L. Feliz**, and ... A tidally tilted sectoral dipole pulsation mode in the eclipsing binary TIC 63328020. *Monthly Notices of the Royal Astronomical Society*, February 2021
21. Joseph E. Rodriguez, Samuel N. Quinn, George Zhou, ..., **Dax L. Feliz**, and ... TESS Delivers Five New Hot Giant Planets Orbiting Bright Stars from the Full-frame Images. *The Astronomical Journal*, 161(4):194, April 2021
22. Joni-Marie C. Cunningham, **Dax L. Feliz**, Don M. Dixon, and ... A KELT-TESS Eclipsing Binary in a Young Triple System Associated with the Local “Stellar String” Theia 301. *The Astrophysical Journal*, 160(4):187, October 2020
23. Romy Rodríguez Martínez, B. Scott Gaudi, Joseph E. Rodriguez, ..., **Dax L. Feliz**, and ... KELT-25 b and KELT-26 b: A Hot Jupiter and a Substellar Companion Transiting Young A Stars Observed by TESS. *The Astronomical Journal*, 160(3):111, September 2020
24. Weicheng Zang, Subo Dong, Andrew Gould, ..., **Dax L. Feliz**, and ... Spitzer + VLTI-GRAVITY Measure the Lens Mass of a Nearby Microlensing Event. *The Astrophysical Journal*, 897(2):180, July 2020

25. Joseph E. Rodriguez, Jason D. Eastman, George Zhou, ..., **Dax L. Feliz** , and ... KELT-24b: A $5M_J$ Planet on a 5.6 day Well-aligned Orbit around the Young $V = 8.3$ F-star HD 93148. *The Astronomical Journal*, 158(5):197, Nov 2019
26. Jonathan Labadie-Bartz, Joseph E. Rodriguez, Keivan G. Stassun, ..., **Dax L. Feliz** , and ... KELT-22Ab: A Massive, Short-Period Hot Jupiter Transiting a Near-solar Twin. *The Astrophysical Journal*, 240(1):13, Jan 2019